

Detailed Experiment Configuration and Results

Experiment	Phase	Epochs	Batch Size	LR	Augment	Valid	R@5	R@10	nDCG@5	nDCG@10
phase1_baseline	phase1	1	16	2e-05	No	No	0.2257	0.3076	0.1961	0.2303
phase1_epochs3	phase1	3	32	2e-05	No	Yes	0.2845	0.3388	0.2462	0.2709
phase1_epochs5	phase1	5	32	2e-05	No	Yes	0.3522	0.4693	0.3176	0.3671
phase2_lr1e5	phase2	3	32	1e-05	No	Yes	0.2694	0.3309	0.2337	0.2605
phase2_lr5e5	phase2	3	32	5e-05	No	Yes	0.3073	0.4208	0.2854	0.3331
phase2_augmentation	phase2	3	32	2e-05	Yes	Yes	0.3868	0.5099	0.3588	0.4098
phase3_hybrid	phase3	3	32	2e-05	No	Yes	0.2845	0.3388	0.2462	0.2709